ParentSync

Your family's WhatsApp and email messages, automatically organized into calendar events

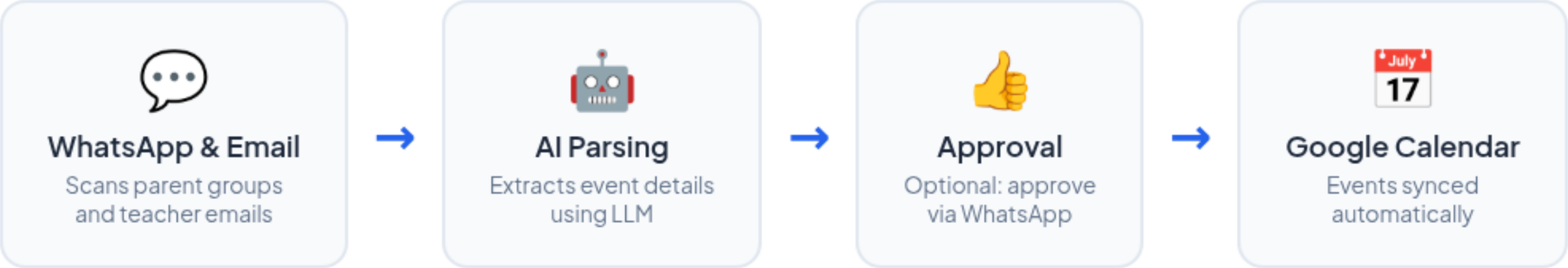
v1.0.1 | Desktop App for Linux, Windows & macOS

What is ParentSync?

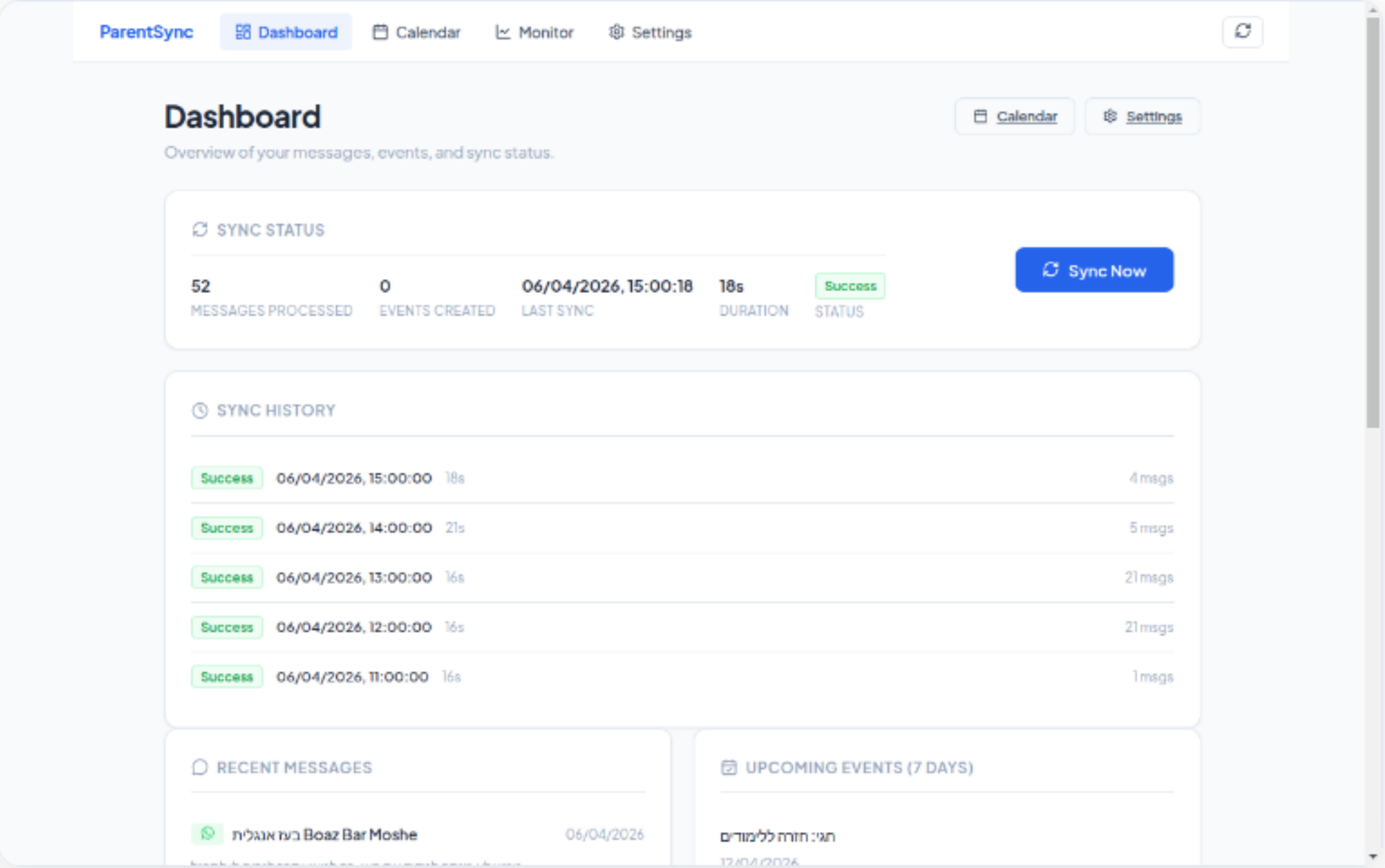
A desktop app that reads your children's school WhatsApp groups and teacher emails, uses AI to find events, and puts them on your Google Calendar.

- Monitors WhatsApp parent groups and Gmail automatically
- AI-powered parsing extracts dates, times, and event details
- Events sync directly to your family Google Calendar
- Optional approval workflow via WhatsApp reactions (👍 approve / ☹️ reject)
- All data stored locally on your machine — no cloud, no sign-up

How It Works

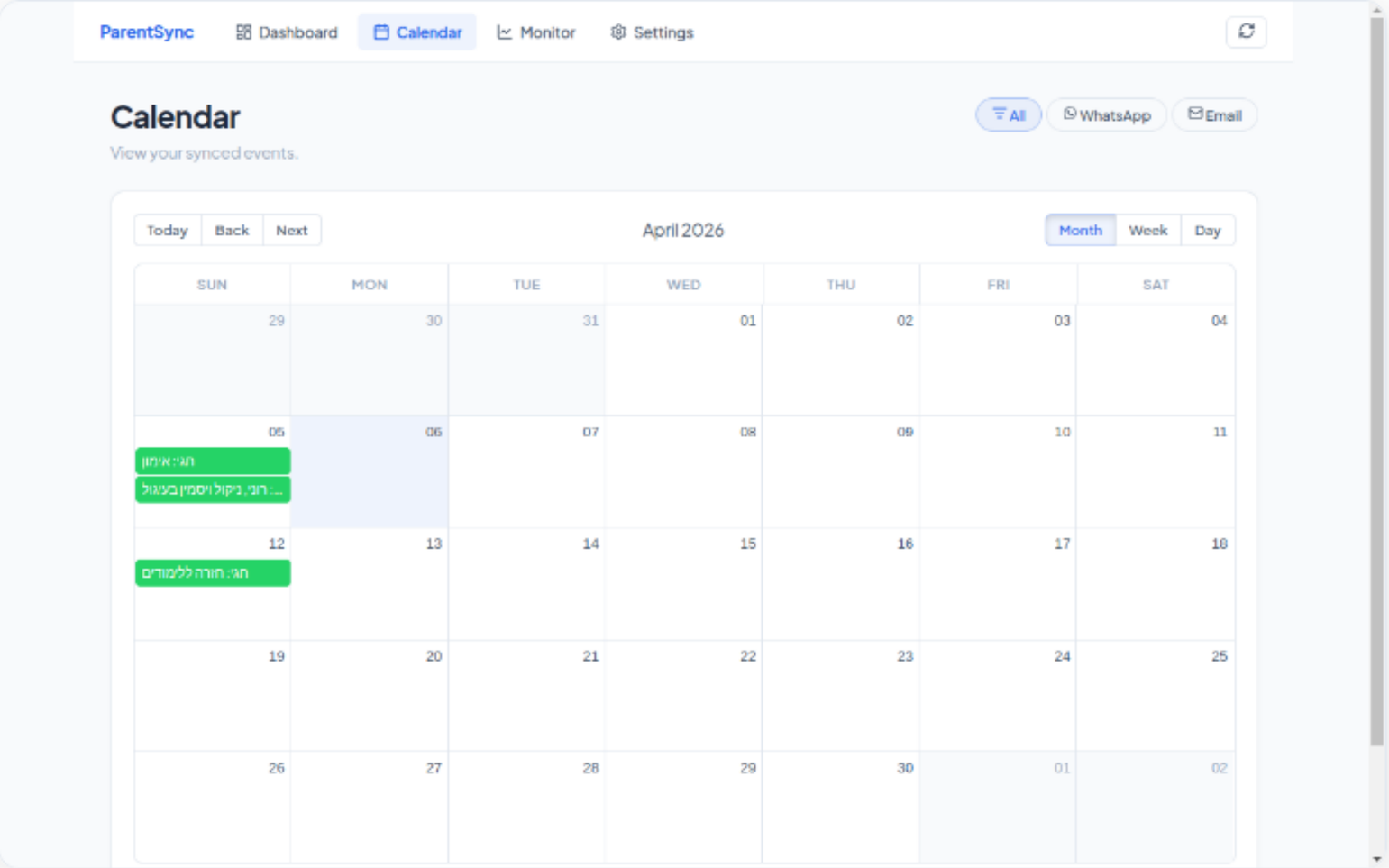

WhatsApp & Email
Scans parent groups and teacher emails
AI Parsing
Extracts event details using LLM
Approval
Optional: approve via WhatsApp
Google Calendar
Events synced automatically

Dashboard



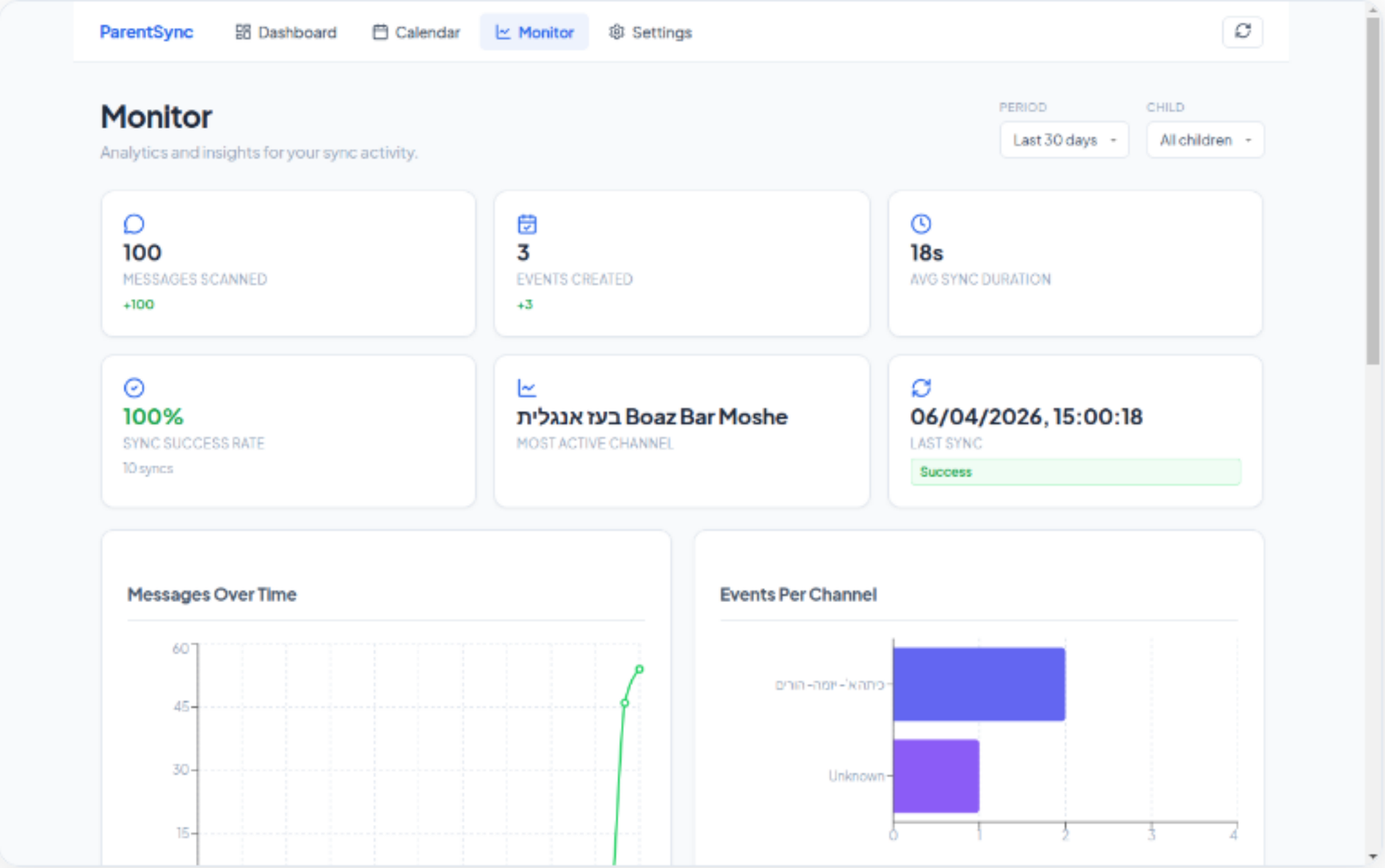
Sync status, recent messages, upcoming events, and sync history at a glance

Calendar View



Full month/week/day calendar with events color-coded by source (WhatsApp or Email)

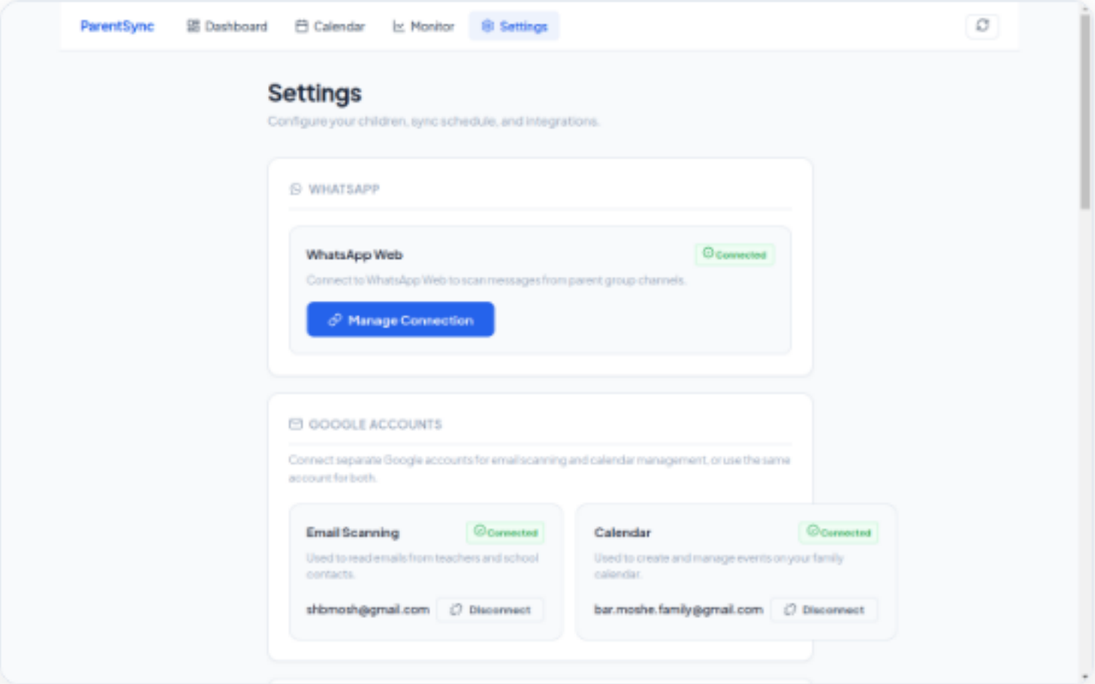
Monitor & Analytics



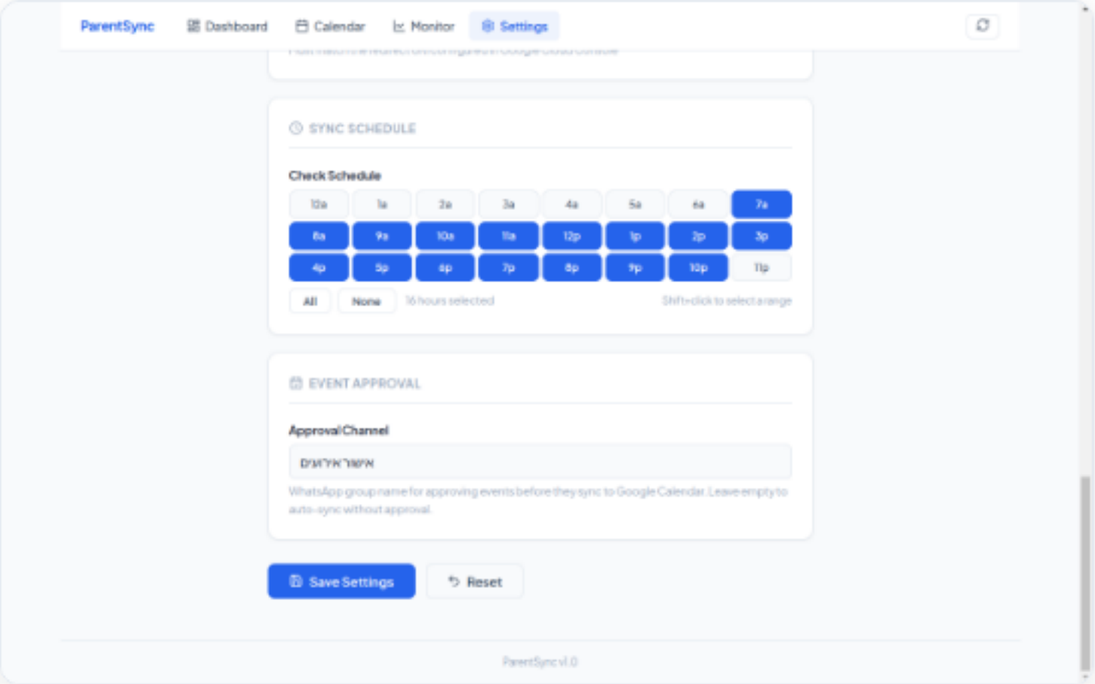
Messages scanned, events created, sync success rate, and channel activity charts

Settings & Setup

Connections



Children & Schedule



Dashboard — In Detail

Sync Status (top card)

- **Messages Processed** — total messages collected across all recent syncs
- **Events Created** — how many calendar events the AI extracted
- **Last Sync** — when the last sync happened
- **Duration** — how long it took (e.g., 18s)
- **Status badge** — green "Success", orange "Partial", or red "Failed"

💡 Click **"Sync Now"** to trigger an immediate scan of all your configured WhatsApp groups and email addresses.

Sync History (expandable list)

- Shows the **last 5 syncs** with status, time, duration, message count
- **Click any entry** to expand — see per-channel breakdown (which child, which group, how many messages)
- Skipped channels show the **reason** (e.g., "no new messages")
- Click **"View synced messages"** to see exact messages that were collected

Dashboard — Messages & Events

Recent Messages (bottom left)

- All messages collected from WhatsApp and Gmail
- Source icon: **WhatsApp icon** = WhatsApp group, **Mail icon** = email
- **Click a message** to expand and read full content
- Click **"Show more"** to load additional messages (20 at a time)

Upcoming Events (bottom right)

- Events in the **next 7 days** parsed from messages
- Shows title, date, time, and location
- **Checkmark icon** = synced to Google Calendar

How to get data flowing

1. Make sure **WhatsApp is connected** (Settings → WhatsApp → "Connected")
2. Add at least one **child with WhatsApp channels** (Settings → Children)
3. Set your **OpenRouter API key** (Settings → OpenRouter)
4. Click **"Sync Now"** on the Dashboard
5. Wait 15–30 seconds — messages and events appear

💡 **Refresh button** (top-right ↻ icon) reloads all data on the current page without triggering a new sync.

Monitor — Summary Cards

Six cards at the top give you a quick health check. Use the **Period** dropdown (7/30/90 days) and **Child** filter to narrow the view.

💬 Messages Scanned

Total WhatsApp + email messages collected. Green trend arrow shows change vs. previous period.

⏱️ Avg Sync Duration

Average time per sync cycle (scanning + parsing). Under 30s is normal.

📊 Most Active Channel

The WhatsApp group or email source that generated the most messages in this period.

📅 Events Created

How many calendar events the AI extracted from messages. Trend arrow compares to previous period.

✅ Sync Success Rate

Percentage of syncs that completed without errors. Green ≥80%, orange ≥50%, red below 50%.

🔄 Last Sync

Timestamp and status badge of the most recent sync. Tells you when data was last refreshed.

Monitor — Charts & Graphs

📈 Messages Over Time

Line chart with two lines:

🟢 **Green** = WhatsApp messages

🔴 **Red** = Email messages

X-axis: dates, Y-axis: message count. Shows daily volume trends — helps spot when groups are most active.

📊 Events Per Channel

Horizontal bar chart showing event count by WhatsApp group name. Each channel gets a different purple shade. Hover to see percentage of total. Identifies which groups generate the most calendar events.

📉 Sync History

Combo chart — bars + line overlay.

Bars: message count per sync (green = success, orange = partial, red = failed)

Line: sync duration in seconds (right axis)

Spot slow syncs or recurring failures at a glance.

🗓️ Channel Activity Heatmap

Grid with channels as rows, dates as columns. Cell color intensity = message count. Darker purple = more messages. Hover any cell for exact count. Reveals which days and channels are busiest.

Common Errors & Solutions

⚠️ "Sync failed"

WhatsApp may be disconnected, or OpenRouter key is missing/expired. Check Settings → WhatsApp status and OpenRouter API key.

⚠️ "0 Events Created" after sync

The AI didn't find any events in the messages. This is normal if messages are just chat. Only messages with dates/times get parsed into events.

⚠️ "Google OAuth not configured"

You haven't entered the Google Client ID and Secret yet. Go to Settings → Google OAuth and follow the setup steps (slide 16).

⚠️ "WhatsApp is not connected"

Session expired. Go to Settings → click "Connect WhatsApp" → scan QR code again with your phone.

⚠️ "Failed to load dashboard data"

Backend may still be starting. Wait a few seconds and click the **refresh button** (↻ top-right). If it persists, restart the app.

⚠️ Events not appearing on Google Calendar

Check: 1) Google Calendar account is connected (Settings), 2) If using approval channel, react 👍 to approve the event in WhatsApp.

Getting & Refreshing Data

Automatic Sync

ParentSync automatically scans at the hours you configured in Settings → Sync Schedule. For example, if you selected 8am–10pm, it scans once per hour during those times.

Manual Sync

Click **"Sync Now"** on the Dashboard to trigger an immediate scan. Useful after adding a new child or channel.

Refresh Display

The **↻ refresh button** (top-right corner of the nav bar) reloads all data on the current page from the database. It does NOT trigger a new sync — it just refreshes the view.

Data Freshness Tips

1. After connecting WhatsApp for the first time, click **"Sync Now"** to populate initial data
2. Add all your children's group names **before** the first sync so all channels are scanned
3. The Monitor page filters (7d/30d/90d) affect all charts — use **30 days** for a good overview
4. If Dashboard shows old data, click **↻ refresh** — if it still doesn't update, click **"Sync Now"**

💡 The app keeps running and syncing even when minimized to the system tray.

How-To Guide

Step-by-step setup for each integration

Setup: OpenRouter (AI)

The AI engine that reads messages and extracts calendar events

What is OpenRouter?

OpenRouter is a gateway to many AI models (GPT, Claude, Llama, etc.) through a single API key. ParentSync uses it to read WhatsApp and email messages and identify dates, events, and details.

💡 Free models are available! The default model (**qwen3.6-plus:free**) costs nothing to use.

How to get your API key

1. Go to **openrouter.ai/keys**
2. Sign up or log in (Google/GitHub)
3. Click **"Create Key"**
4. Copy the key (starts with `sk-or-`)
5. Paste in ParentSync → Settings → OpenRouter → API Key

Setup: Google Account

Connect Gmail and Google Calendar to read emails and sync events

What is Google OAuth?

OAuth lets ParentSync access your Gmail and Calendar **without storing your Google password**. You create a "Client ID" in Google Cloud Console — think of it as a permission slip that tells Google "this app is allowed to access my data."

💡 You can use **separate Google accounts** for email scanning and calendar — or the same account for both.

How to set it up

1. Go to **console.cloud.google.com**
   Create a project if you don't have one
2. Enable **Gmail API** and **Calendar API**
   APIs & Services → Enable APIs
3. Create **OAuth consent screen**
   Add your email as a Test user
4. Create **OAuth Client ID** (Web app)
   Redirect URI:
   `http://localhost:3000/api/auth/google/callback`
5. Copy **Client ID** and **Client Secret** into ParentSync Settings

Connecting Google Accounts

After entering Client ID & Secret

1. Click **"Save Settings"**
2. Click **"Sign in with Google"** under Email Scanning
3. Google's consent screen opens in your browser
4. Select your account and click **"Allow"**
5. You're redirected back — status shows **Connected**
6. Repeat for **Calendar** (same or different account)

Troubleshooting

⚠️ **"Error 403: access_denied"**

Your Google app is in "Testing" mode. Fix: Go to OAuth consent screen → either **add your email as a Test user** or click **"Publish App"**.

⚠️ **"Failed to exchange authorization code"**

The Client Secret may be wrong, or the redirect URI doesn't match. Double-check both in Google Cloud Console and in ParentSync Settings.

Setup: WhatsApp

Connect your WhatsApp to scan parent group messages

How it works

ParentSync connects to WhatsApp Web — the same way you'd use WhatsApp on your computer. It runs its own browser in the background to read messages from the groups you specify.

💡 Your WhatsApp session **persists across app restarts**. You only need to scan the QR code once. If the session expires, you'll be prompted to re-scan.

📱 Your phone must stay connected to the internet for WhatsApp Web to work (WhatsApp requirement).

How to connect

1. Open ParentSync → **Settings**
2. Click **"Connect WhatsApp"**
3. A **QR code** appears in the app
4. On your phone:
   WhatsApp → Settings → Linked Devices → Link a Device
5. Scan the QR code with your phone camera
6. Status changes to **"Connected"**

Then go to **Children** in Settings, add a child, and type the exact WhatsApp group name.

Under the Hood

DESKTOP SHELL
Electron (single executable)

BACKEND
NestJS (REST API, embedded)

AI
OpenRouter (multiple LLM models)

FRONTEND
React + TypeScript

DATABASE
SQLite (local, no server)

WHATSAPP
whatsapp-web.js (Chromium)

All data stored locally on your machine. No cloud servers, no accounts to create. Just download, configure, and run.

Platform Support

Platform	Format	How to Run
Linux	.AppImage (portable)	chmod +x, double-click or run from terminal
Linux	.deb (Debian/Ubuntu)	dpkg -i, run from app menu
Windows	.exe (installer)	Double-click installer
macOS	.dmg	Drag to Applications

```
# Linux quick start
chmod +x ParentSync-1.0.1.AppImage
./ParentSync-1.0.1.AppImage
```

Built with Electron | All data stored locally | Private use

Event Reminders

ParentSync sends a WhatsApp reminder ~24 hours before an event begins, but only for events that have settled in the system long enough to be trusted.

- Hourly cron scans the calendar for upcoming events
- Only events added more than **24 hours ago** are eligible — protects against last-minute parsing mistakes
- Only events already **synced to Google Calendar** qualify
- Event existence is **verified against Google Calendar** before sending — deleted events are silently skipped
- Reminder is delivered to the configured approval/notification WhatsApp chat with full details: title, date, time, location, description, and source channel
- Each event is reminded at most once (`reminderSent` flag)

See `docs/EVENT-REMINDERS.md` for the full specification.